Abdullah Al Zahid

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SUMMARY

Transportation engineer with a master's degree whose work has centred on data and models: Python tooling that summarised large GO operations files at Metrolinx, simulation of GO service changes, and travel data analysis and dashboards at the City of Toronto. Turns analysis into clear recommendations for decision-makers.

EXPERIENCE

Assistant Planner, Transportation Planning

City of Toronto | January 2024 – present

- Analyze travel and network data, including several years of the Transportation Tomorrow Survey, to establish where capacity falls short and what that means for future service and investment.
- Built the Transportation Data Dashboard, replacing a report rebuilt by hand each quarter with one validated source the division reports from, and saw the change through to adoption.
- Build and maintain the road and right-of-way datasets the division depends on, using GIS and Python, including a ROW-width dataset covering all city roads, built with QGIS, MapBox and JavaScript.
- Run assigned studies from scoping through to recommendations, writing the briefing notes that come out of them and presenting to planners, engineers and managers in other divisions.

Transportation Planner, Simulation and Modelling

Metrolinx, Service Design | February 2023 – January 2024

- Built the Crew Variance Tool, a Python application that read large JSON operations files and produced summaries of speed profiles and run-time variance across subdivisions.
- Set up a command line workflow so temporary and permanent slow orders could be pre-simulated in batches, cutting turnaround on scenario requests from days to hours.
- Ran OpenTrack simulations of GO rail services to establish what proposed operating and infrastructure changes would do to run times, reliability and recovery before they were taken forward.
- Worked with schedulers, operations staff and engineers on service design decisions and presented the evidence behind them.

Graduate Transportation Planner

Mott MacDonald | January – February 2023

- Short consulting placement on Ontario Line work. Ran traffic simulations to estimate how construction staging would affect surrounding streets, and prepared the supporting material for the project team.

Graduate Research Assistant

University of Alberta | January 2020 – August 2022

- Master's research on how road networks perform during emergency evacuations, using macroscopic (LWR) and agent-based (MATSim) models.
- Worked with large travel datasets and wrote up results for readers without a modelling background.

SKILLS

Python (Pandas, NumPy) and SQL for data extraction, analysis and automation. Simulation and modelling: OpenTrack, MATSim, VISSIM and Synchro. Advanced Excel, including large multi-source workbooks. Dashboards and data validation. ArcGIS Pro and QGIS; JavaScript. Reports and presentations for senior staff. MS Office.

EDUCATION

MSc, Transportation Engineering — University of Alberta, 2022. CGPA 4.0/4.0; fully funded by the university.

BSc, Civil Engineering — Bangladesh University of Engineering and Technology, 2018. CGPA 3.69/4.0; University Merit Scholarship, University Stipend Scholarship, Technical Scholarship, Dean's List.

BSc, Computer Science — University of British Columbia, in progress (2025–2027)